Microsoft

Managed Operations: Build, Deploy and Operate at enterprise scale

Casey Burke

Principal Product Manager, Microsoft

Greg Lindhorst

Principal PM Architect, Microsoft

Arjun Mayur

Product Manager II, Microsoft

Christopher Robbins

Director of Application Development, Lummus Technology

MS38 | Tuesday | 11:30AM

Agenda

- Overview
- Power Platform ALM
- Customer spotlight: Lummus Technology
- Copilot Managed Runtime ALM
  - Projects and Stages
  - GitHub integration
  - Deployments and Approvals
- Monitor
- Intelligent, autonomous operations

AI changes how we build.
And what we're able to build.

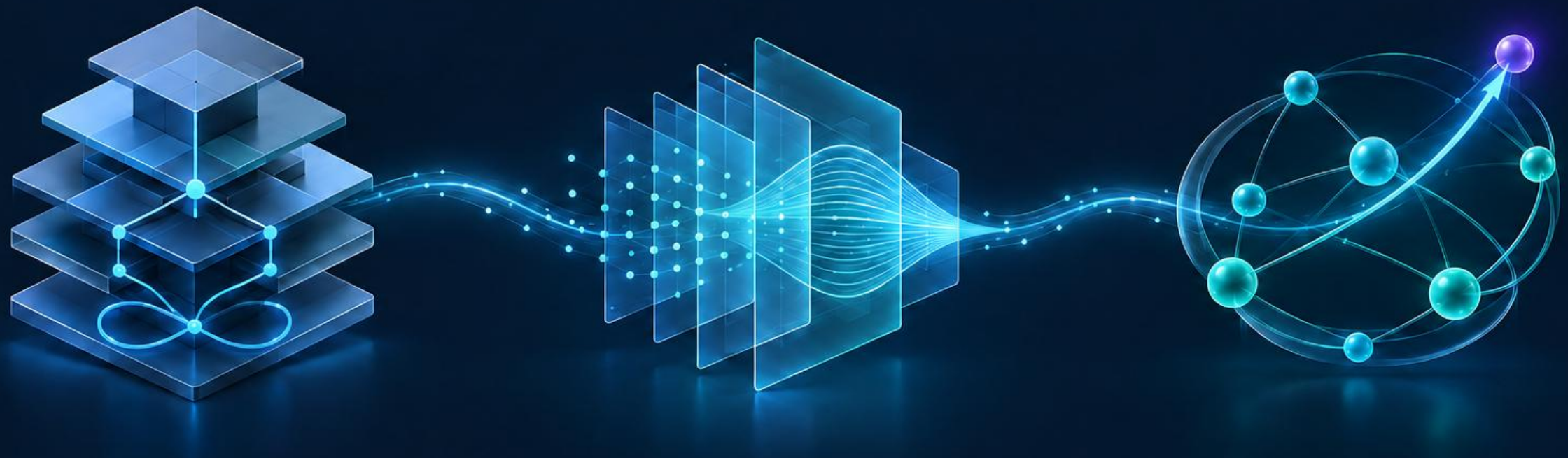
How can we securely and reliably operate **100x** more applications?



Agentic development requires agentic operations



The future of DevOps



ESTABLISHED

Foundation

Git · Deploy · Extend · Recover · Monitor

Intelligence

Signals into insight

Quality · Reliability · Efficiency

NEXT FRONTIER

Autonomy

Specialized AI agents

Autonomous. Self-improving.

Give agents and humans the tools to act

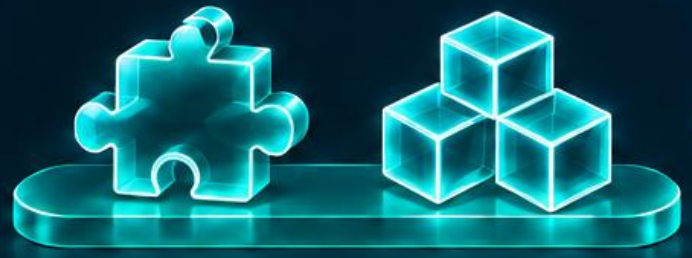
Expose existing tools

Deterministic operations



Add new tools

New capabilities



Agent-ready interfaces





Agents

CMR adds a tool. Not a replacement.



Power Platform ALM

Dataverse +
solution-aware assets

Power Apps code apps, too

+

FIRST LOOK



Copilot Managed Runtime ALM

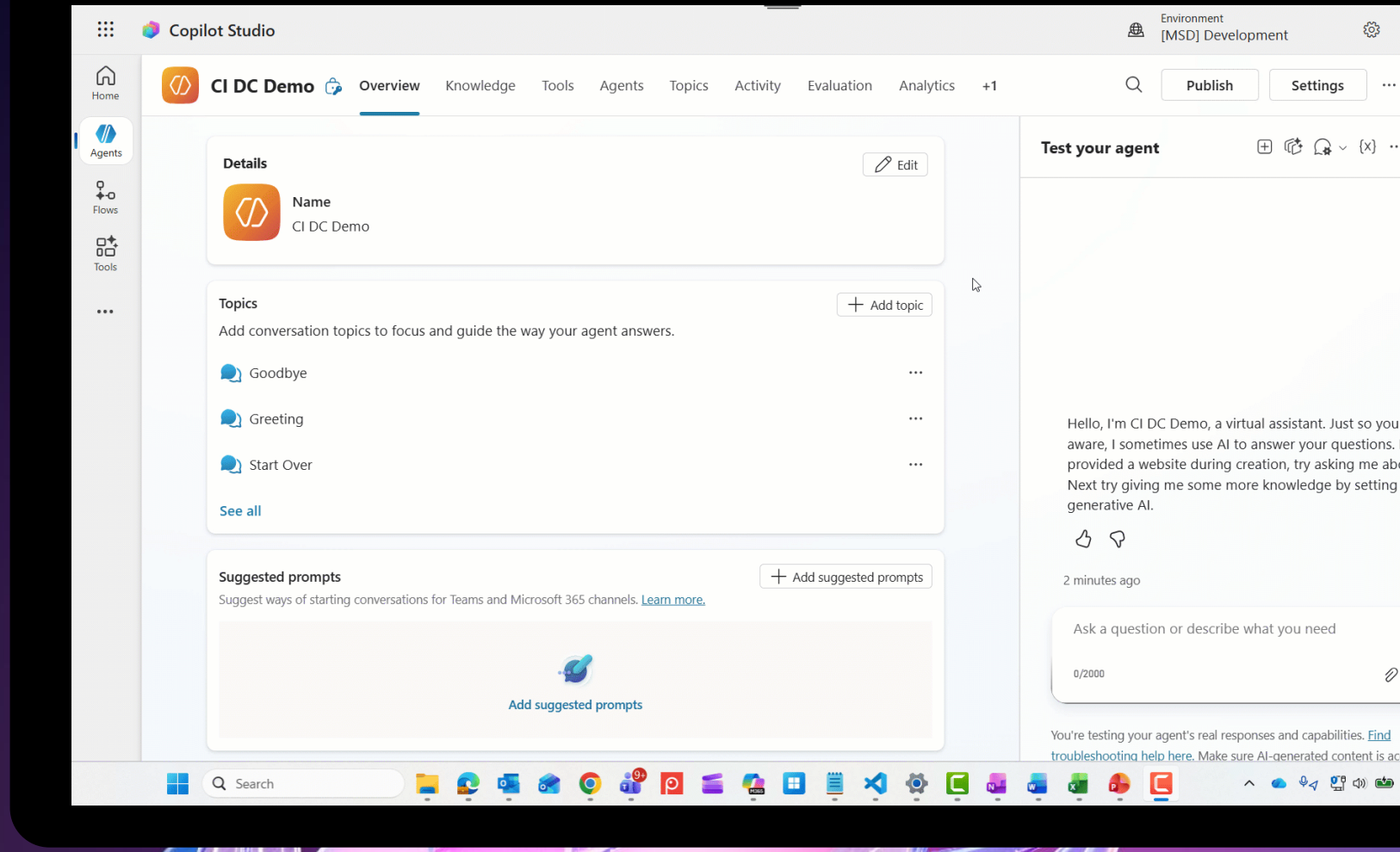
Git-based ALM for CMR-hosted projects

Humans define the structure and expose tools to agents

Pipelines in Power Platform

Generally Available

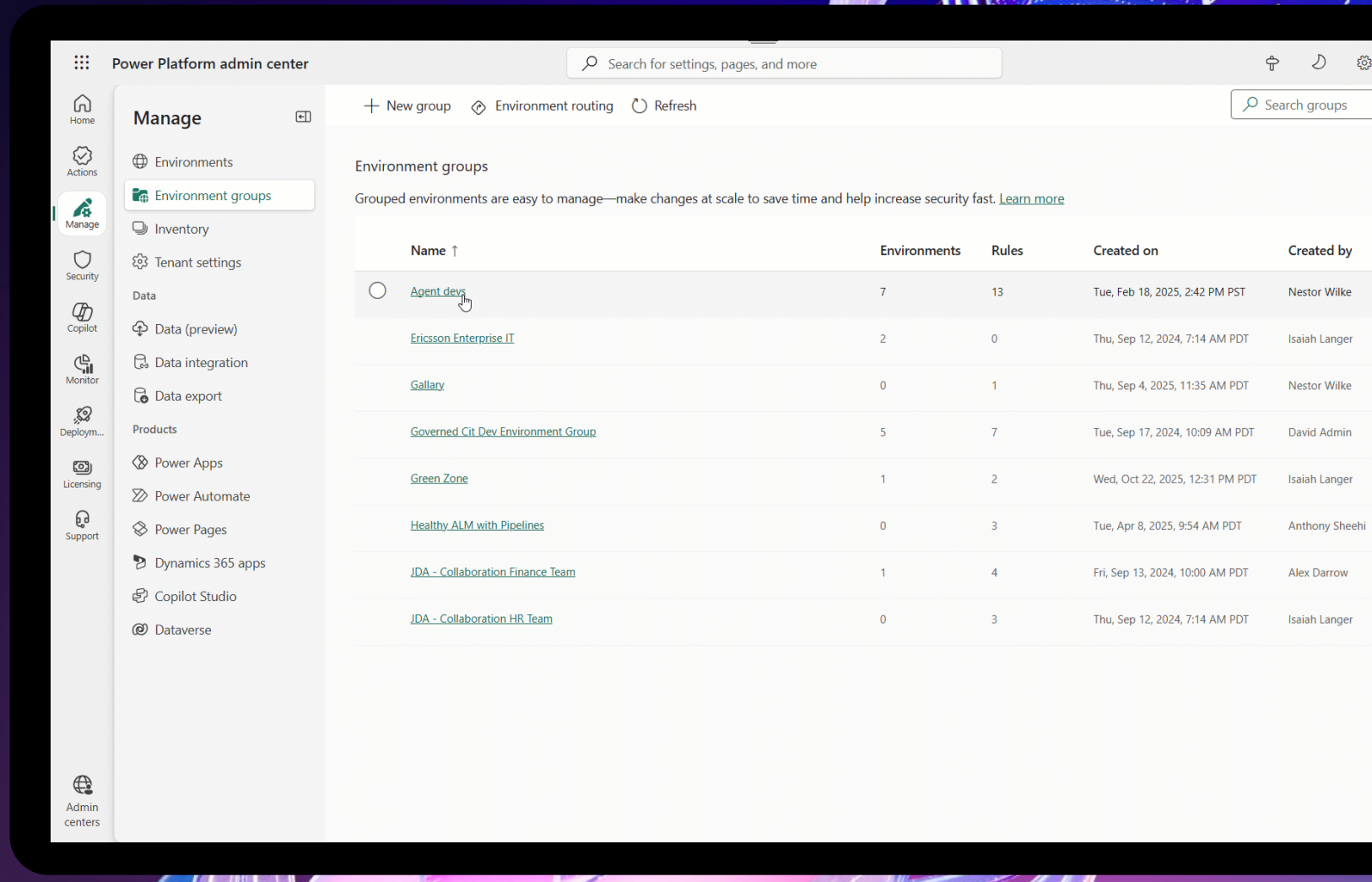
- Simple, powerful, secure
- Enforced best practices help makers succeed
- Securely deploy w/ approvals
- Admins have full visibility in the Deployment hub
- Available in all Power Platform maker portals, PPAC, and **PAC CLI**



Default pipeline in Environment groups

Generally Available

- Admins configure the deployment process once
- Maker environments are automatically enabled
- Managed per Environment **group**
- Recommended with environment routing and sharing limits



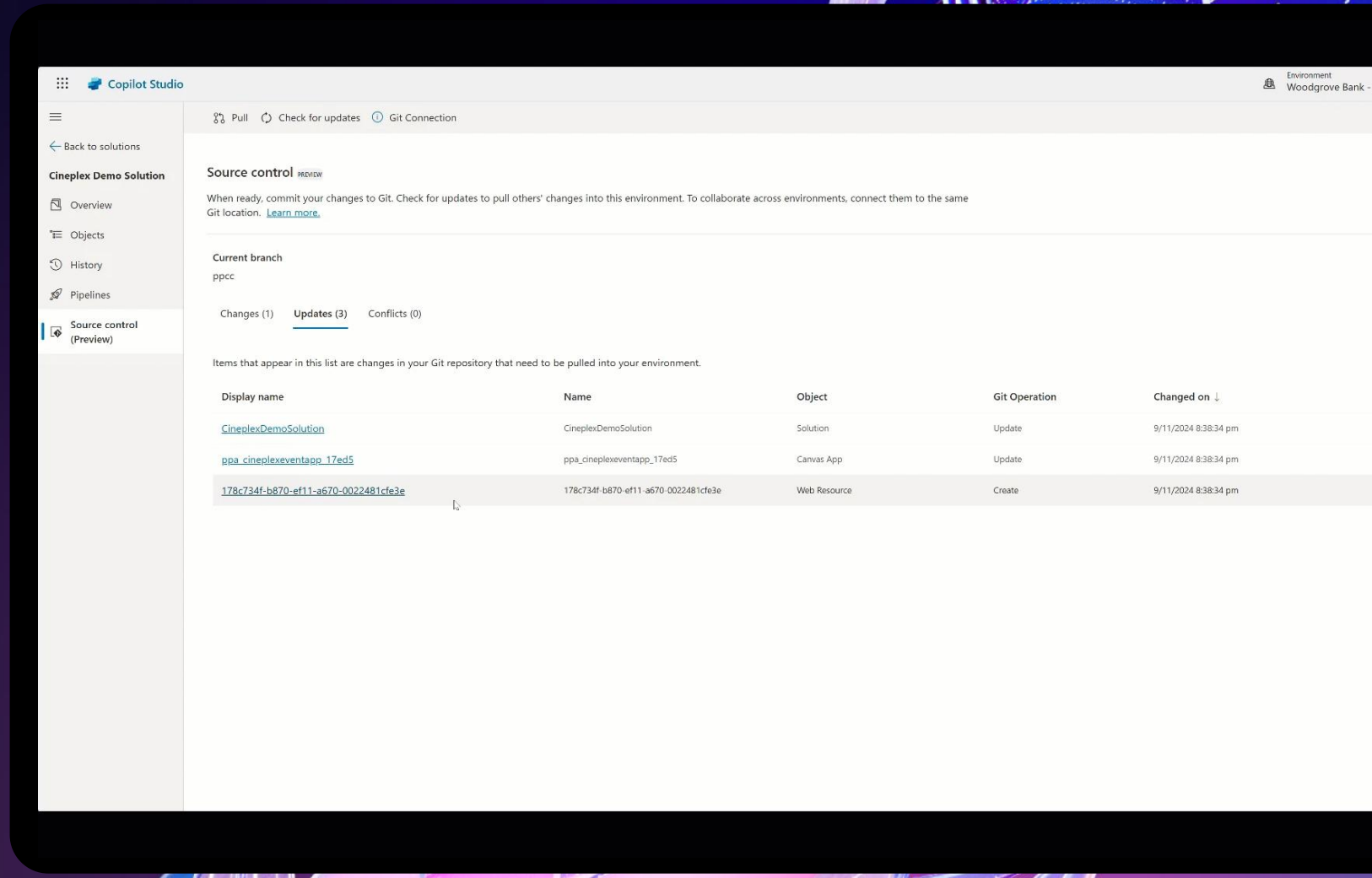
Native Git Integration

Generally Available

- Push and pull changes between dev environments and Git
- Faster than classic tools
- New YAML format is easier to read and merge
- Supported in PAC CLI and has public API's

Supports GitHub (preview)!

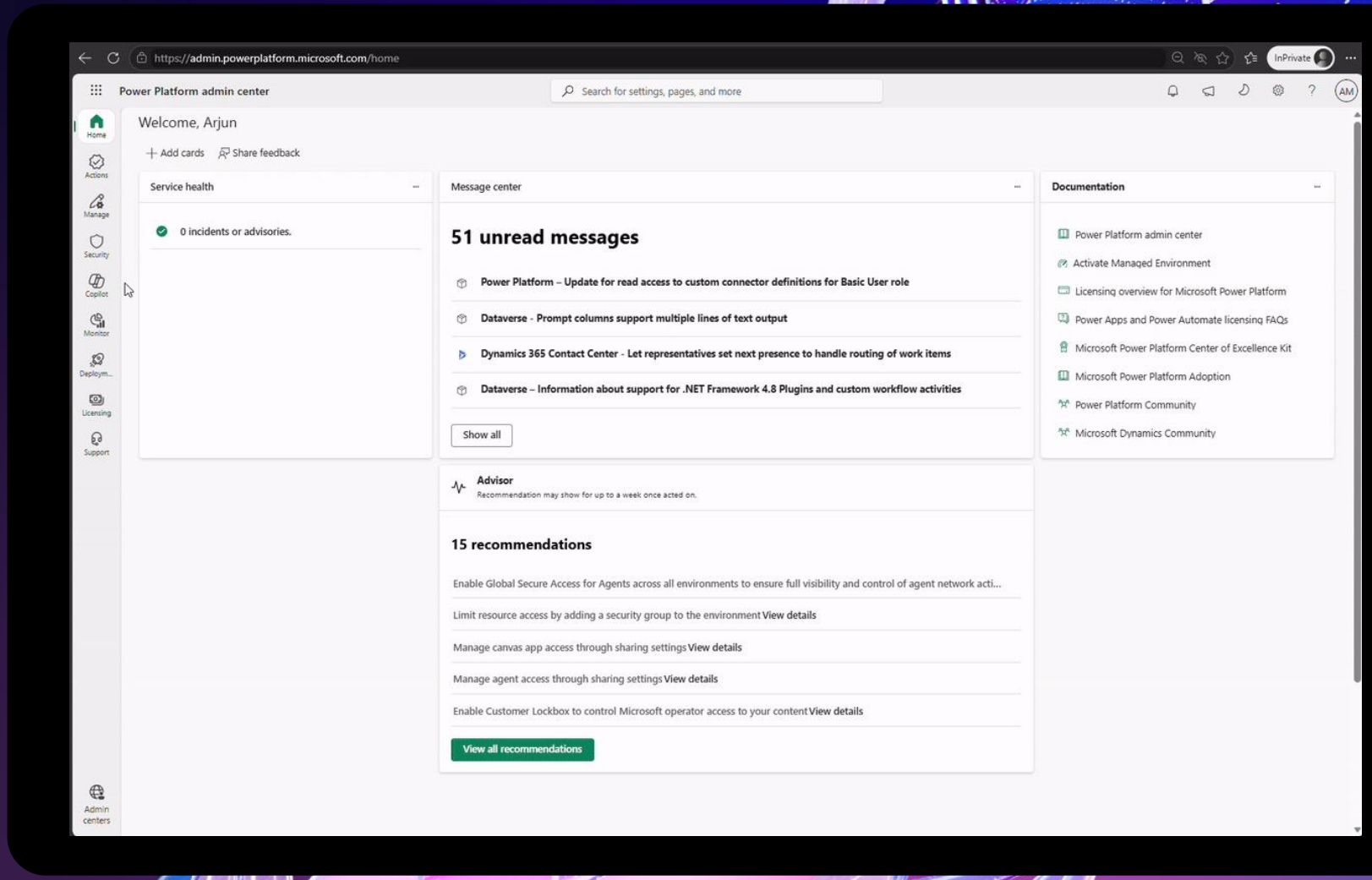
aka.ms/SourceCodeIntegration



Monitor the health of your portfolio

Generally Available

- Monitor provides metrics, recommendations and alerts to stay on top of operational health
- Today, it supports these for apps, agents, automations (and now sites!)
- Available today in PPAC, MAC, and all maker experiences (Automation Center in PAuto)



aka.ms/MonitoringHub

Lummus Technology

Christopher Robbins

Director of Application Development

From code change to customer-ready release

Agents help us move faster; approvals and observability keep us in control.

Agent or automation step Human review / decision

01 BUILD & REVIEW

Agents assist with development, testing, and review

Engineers review and approve the change

02 RELEASE & ENABLE

Release automation publishes the Power Apps code app

REVIEWABLE DRAFTS

A deployment event starts drafts for release notes, training/release videos, and SharePoint pages

An owner reviews and publishes the content

03 OPERATE & LEARN

Application Insights monitors the app and our Azure Functions engineering calculations

Telemetry helps the team spot and investigate issues

An agent can help summarize findings or propose a fix, which goes back through review

FOUNDATION: Source control · Gated deployments · Human approval · Shared telemetry

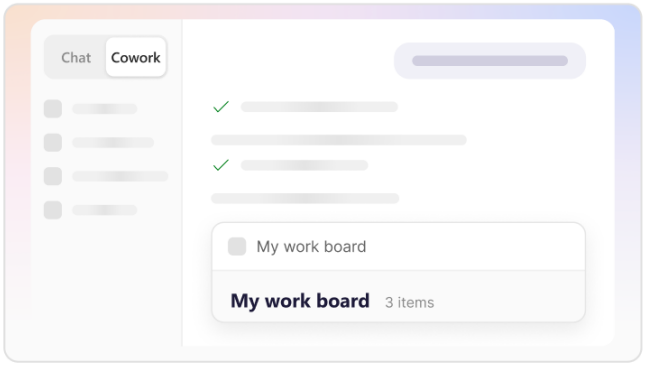
Show of hands:

Have you used the Copilot Managed Runtime??

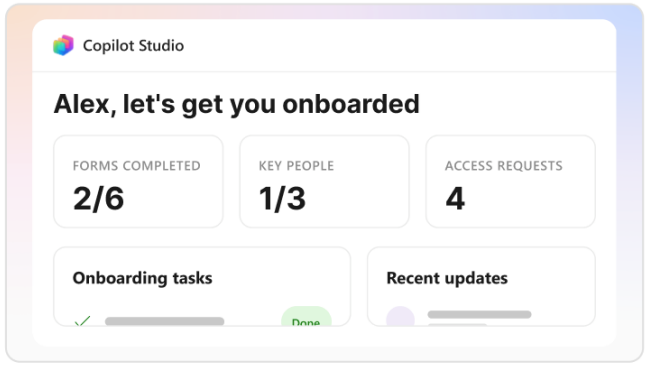
Microsoft entry points

There are four Copilot entry points for making artifacts and hosting with Copilot Managed Runtime

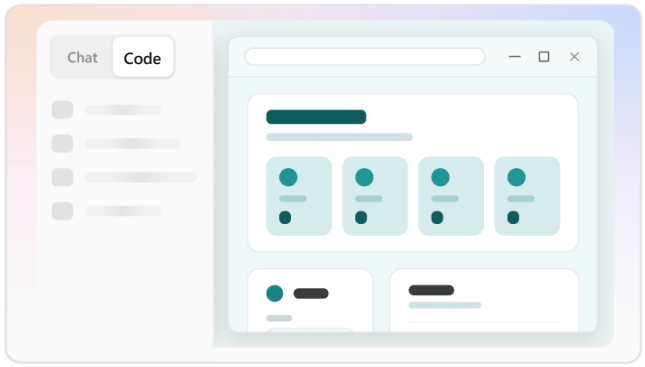
Copilot Managed Runtime host



Copilot Cowork



Copilot Studio



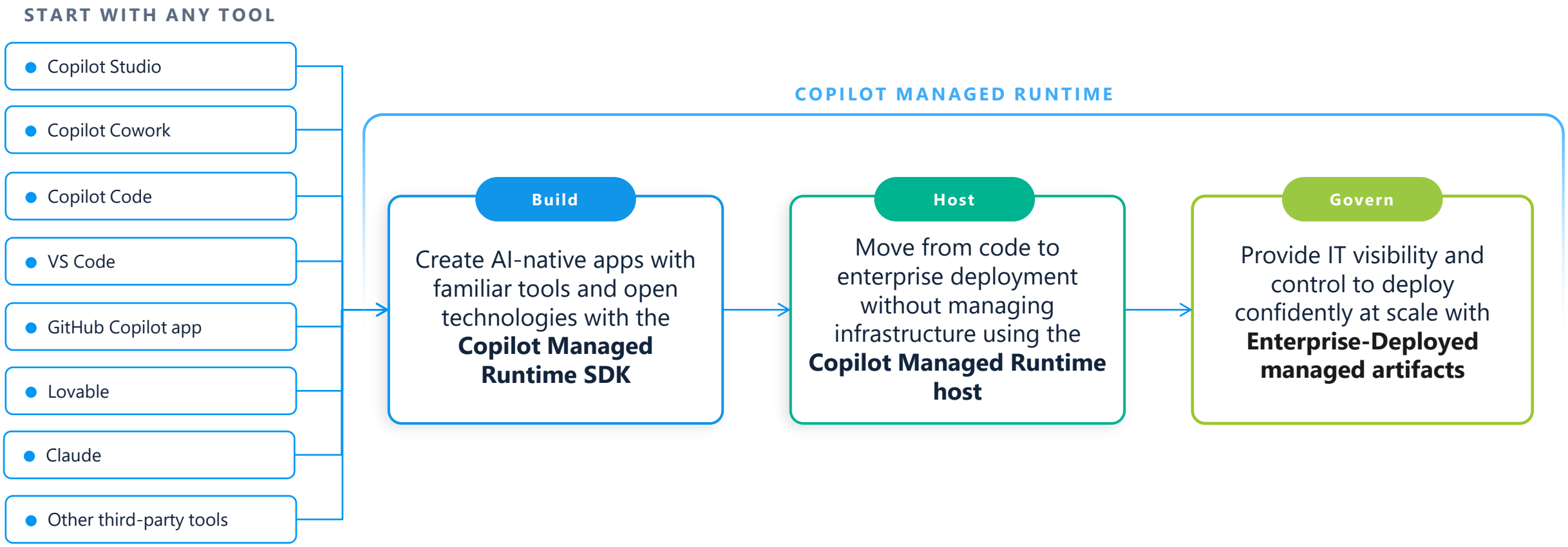
Copilot Code



GitHub Copilot

The Copilot Managed Runtime platform

Connecting your platform and enterprise deployment. Build on any platform. The SDK enables a seamless handoff to the Host and enterprise deployment.



Start anywhere — the SDK connects — artifacts become hosted → an enterprise deployed artifact

Enterprise ready from the start

Apps hosted with Copilot Managed Runtime are enterprise ready with complete IT controls built in from the start



Enterprise management, governance, and protection are built in.

Build enterprise ready applications with the management, governance and protection built in.

ALM aware from the start

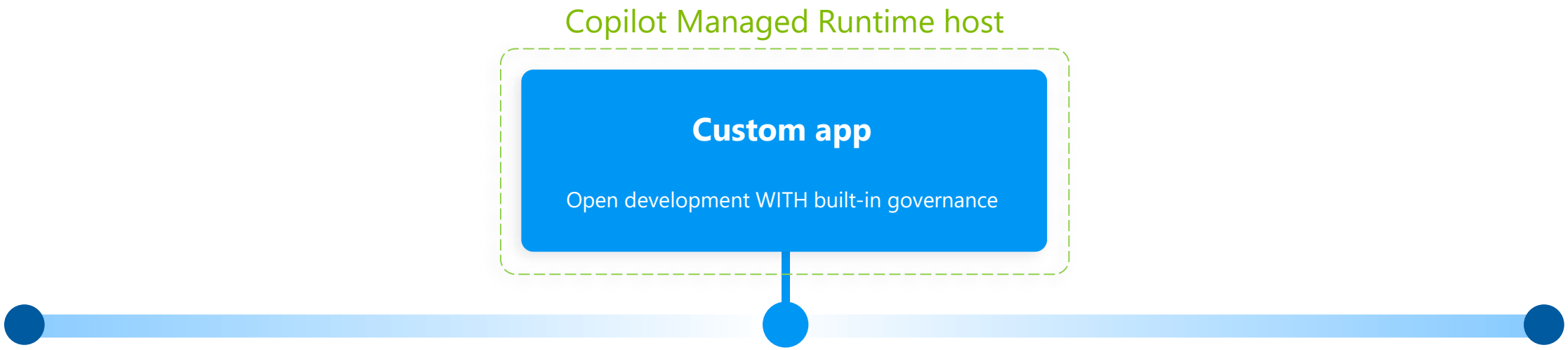
From solutions and environments to Git-backed projects and stages

Power Platform ALM	Copilot Managed Runtime ALM	Maker gains
Pick your environment or use the shared Default environment	Default to individual personal environments	**LESS DECISIONS** Environment is one less thing to understand and pick
Dataverse solutions: layers + dependencies	Always backed by a Git repo, possibly GitHub; no Dataverse required for ALM	**LESS TO LEARN** Skip solution mechanics, build in your IDE of choice
Configure pipelines; add Git and testing	Repo and build/deploy wired together; BYO tools optional	**LESS TO SET UP** Built-in ALM
Move solutions across Dev / Test / Prod environments	Deploy commits through stages; preview, then explicitly go live with approvals	**CONTROL RELEASES** Choose when to go live

Environments remain governance boundaries. Configuration lives in projects with stage overrides.

Between opinionated and open

An app hosted on Microsoft Copilot Managed Runtime uses the best of both for your developed apps



Projects and Stages

Three shifts from the ALM we know today

The unit of work, where stages live, and what a personal environment is allowed to host.

Git Backed Projects

A project is a Git repo holding the apps, agents, and data that build and deploy together. Git repo can be in GitHub.

Single and Multi Environment Deployment

Every environment can host test and production stages on its own, as well as deploying to other environments if needed.

Personal Environment can host Prod

A personal environment can host a production app for the long term.

Single environment is the default path; multi-environment ALM stays available for advanced cases.

A project is one repo that builds and deploys as a unit

Public Preview ETA:
End of November

PROJECT = REPO

```
my-project/
   ms.project.config.json
   driver-app/
      ms.config.json
   dispatcher-app/
      ms.config.json
```

One directory per component, each with a config file.

We will continue to list apps independent of projects – makers and end users do not need to think about projects.

Select portions of config files can be overridden for a specific deployment.

Projects support three ALM "stages"

Public Preview ETA:
End of November





APP AUTHORS

Every commit produces a runnable version, for authors only.



APP TESTERS

Optional. Multiple deployments for a small, invited audience.



END USERS

The single live production deployment, for a broad set of users.

Deployments

- Run local build
- Run cloud preview

- Deploy to prod, the default target
- Create additional targets and deploy
- Run from a target with user permissions

Three patterns, from simple to advanced

We expect the vast majority of apps to never leave the environment they were built in.

~80%

STAGES IN A PERSONAL ENV

DEV ✓ TEST ★ PROD

An idea becomes a real app for real users, all in the maker's own environment.

As long as the project can live within the governance constraints of the Personal Environment, it need not be ever deployed elsewhere.

~15%

STAGES IN A SHARED ENV

DEV ✓ TEST ★ PROD

A team development environment with its own policy, still holding all three stages together.

Public Preview ETA: End of November

~5%

MULTI-ENVIRONMENT ALM



Production runs in its own environment, for the governance controls that need that boundary.

Public Preview ETA: End of Year

Using a single environment removes friction

An idea that starts in a Personal Environment can be tested with a small audience, then deployed broadly - same environment, same controls.

CONSISTENT CONNECTOR SETUP

Connections are created once and stay put. Nothing has to be recreated, or re-permissioned, in another environment.

FAST AND RELIABLE

Promotion takes seconds rather than minutes, with no cross-environment move that can fail part way.

SAME GOVERNANCE CONTROLS

Policy is set per environment, so dev, test, and production inherit exactly the same controls.

NO ADMIN ACTION REQUIRED

Makers wait for nothing to be provisioned. The road and the destination are already there.

Deployment target configuration overrides

Each target can override connection references, project variable values, and other configuration values

Public Preview ETA: End of November

PROJECT = REPO

```
my-project/
    ms.project.config.json
    ms.project.prod.config.json
    driver-app/
        ms.config.json
        ms.prod.config.json
        ms.test.config.json
    dispatcher-app/
        ms.config.json
```

Project variables will be introduced, replacing Environment variables. Overridable at the project level.

Connection references can be overridden for each component.

Additional deployment targets get their own override files.

If overrides are not provided for a target, the base is used.

ALM Summary

From environment- and solution-centric deployment to project-, Git-, and stage-centric lifecycle management

	Traditional Power Platform ALM	Copilot Managed Runtime ALM
Primary unit	Dataverse solution	Git-backed project
Lifecycle topology	Dev → Test → Prod environments	Dev → Test → Prod stages
Source control	Optional or added to the process	Native and expected
Deployment motion	Export / import solutions	Deploy specific project commit
Environment role	Lifecycle + governance boundary	Governance boundary
Configuration	Environment variables + connection references	Project configuration + target overrides
Dataverse relationship	Core ALM foundation	Independent; can coexist
Pipeline role	Core deployment mechanism	Optional orchestration / governance

Bottom line: traditional ALM moves solutions between environments; Managed Apps ALM moves project versions through stages.

GitHub Integration, Deployment, and Approval

Every Project has a Git Repo

Public Preview

Projects support all repo types

OOB repos are the default

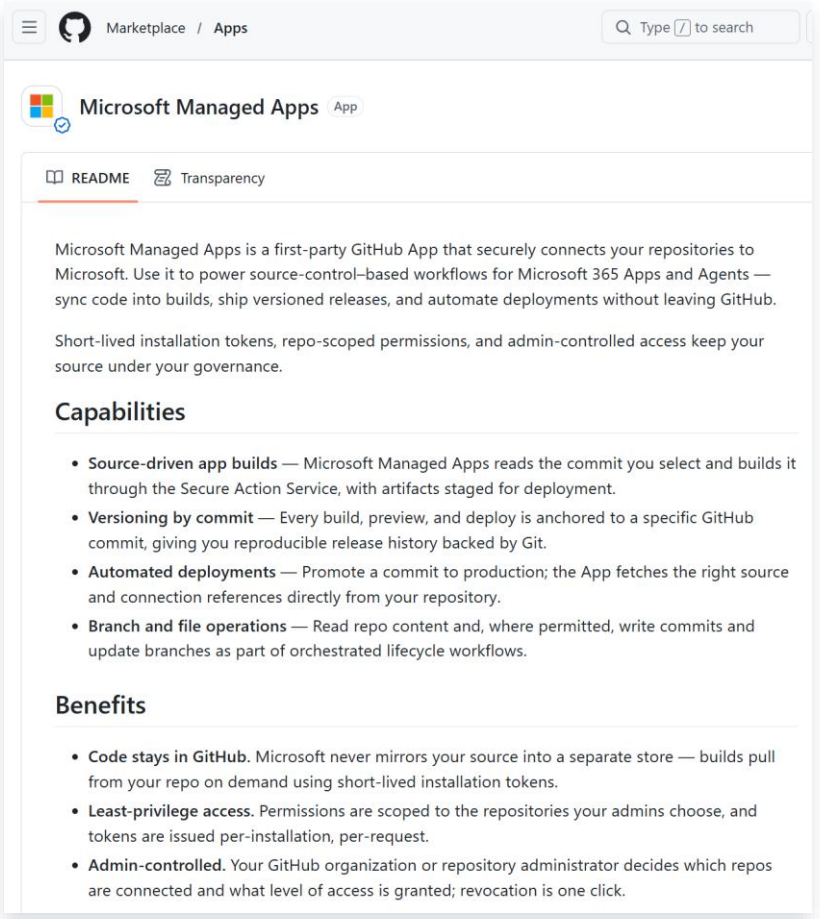
- Everything is version controlled within a remote Git repo (can be cloned locally)
- Source code is your single source of truth for development. Not solutions; not a Dataverse DB

GitHub Enterprise native integration

- Optionally host code in your own remote GitHub repos
- It's automatically wired into the platform's build-and-deploy pipeline

Bring-Your-Own

- Want more flexibility? Manage the E2E build and deployment with your Git system of record
- Supports service principal (SPN) auth for autonomous workflows



CMR Deployment and Approval

Coming soon

ROLES

PROJECT OWNER

Turns on approval by assigning approvers to a target

```
ms project add approval <email>
```

Approval is required while at least one approver is assigned.

CONTRIBUTOR OR AGENT

Requests the deployment

```
ms project deploy
```

Can deploy, but can't approve unless explicitly assigned by the owner or admin.

APPROVER

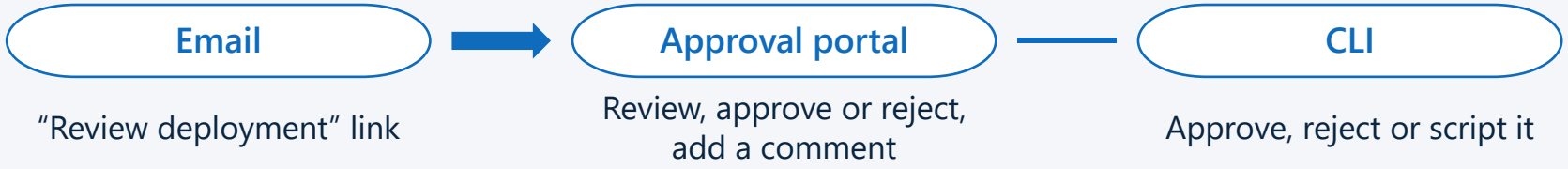
Reviews the details and decides

```
ms deployment approve
```

Sees what's being deployed, without needing edit rights.

Approve deployments using the CLI or in-product experience.

Works for all repo types



Demo CMR Deployment and Approvals

Monitor

Show of hands

Monitor — operate with confidence



Have you used Monitor?



Have you set up alerts in your tenant?

Operate with confidence

Monitor gives IT and operations teams visibility into Copilot Managed Runtime (Microsoft admin center) and Power Platform (PPAC) health.



Understand health

Operational health metrics provide a digestible way to understand health



Improve health

Actionable recommendations suggest how you can improve health when its degraded



Act proactively

Create alerts with custom thresholds to proactively monitor health

Customer pain points with Monitor



Why wait until tomorrow for alert notifications?



Can alerts reach the workspaces we operate in?



Can we build health dashboards with Monitor metrics?



What about Power Pages?

Detect issues sooner

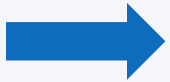
Public Preview ETA: Mid-November

"Why wait until tomorrow?"

ALERT EVALUATION CADENCE

Daily

24 hours



Hourly

1 hour

Today: Copilot Managed Runtime Apps · Power Apps · Cloud flows · Agents

Next: more supported metrics and products

LIVE DEMO

Configure an hourly alert

Metric → Threshold → Hourly rule

Arjun Mayur

Route alerts to where teams work

Public Preview ETA: End of November

"Can alerts reach workspaces we operate in?"

ONE POWER AUTOMATE / LOGIC APP / AGENTIC FLOW



Trigger: every alert in your scope

Extend to any connector-supported destination for both CMR and Power Plat assets.

Home
Actions
Manage
Security
Copilot
Monitor
Deploym...
Licensing
Support
Admin centers

Monitor

Overview
Alerts
Logs

Products

Power Apps
Power Automate
Power Pages
Copilot Studio

Alerts

Get the info you need when you need it most. Configure up to 50 custom alert rules per tenant and we'll notify you when something triggers them. Alerts are a feature of managed environments. Learn more

+ Alert rule

Alert rules | Triggered alerts

Items: 80

Status	Alert rule	Scope	Product
On	[redacted]	Resource	Power
On	[redacted]	Resource	Power
On	[redacted]	Resource	Power
On	[redacted]	Resource	Power
On	[redacted]	Resource	Copilo
On	[redacted]	Resource	Power
On	[redacted]	Environment	Power
On	[redacted]	Resource	Copilo

Take Monitor metrics further

Public Preview ETA: End of November

"Can we build our own health dashboards with Monitor metrics?"

EXPORT MONITOR METRIC AGGREGATES FOR COPILOT MANAGED RUNTIME APPS AND POWER PLATFORM ASSETS

Monitor
Aggregate metrics

↓

Call REST API / PAC CLI
Data export

↓

OpenTelemetry
Aggregate format

↓

Downstream
Dashboards + shared reporting

Retain data longer

Beyond Monitor's 30 days

Join your data

Build custom views

Share insights

Include non-admin team members

```
$profileName = 'Export-Crm'
$uri =
'https://<envid>.environment.api.powerplatform.com/monitoringhub/resourceT
ype/PowerApp/resourceSubType/CanvasApp/exports/metrics?api-version=1'

$body = @{
    resourceIds = @(
        'a714afb8-4ee5-4178-ac71-f1efbef7e387'
        'a68bb328-5dd2-43f2-9011-1231baa0e122'
        '35f0330d-9b73-403b-8fa0-8e3d6dee9dcd'
        '74f0a28d-40e2-446c-9d00-163695ac84f7'
        '1ee15b77-0588-423a-9829-d5c224bed45f'
        'a0389860-15cb-43ef-9af1-ae1bab531f38'
        '76e07991-4f4a-4dec-8d1a-dc0765e93402'
        '74942321-fefb-4fd2-8f1d-907e836ad02e'
        '218bbd25-3f37-4366-9390-28a5ee8652a2'
        '9d0bc1c4-bf5a-4fd6-b3fd-7ec08ab2da2b'
    )
    timeRange = @{
        startDate = '2026-09-21'
        endDate   = '2026-10-14'
    }
} | ConvertTo-Json -Depth 5
```

Power Pages site health

Public Preview ETA: End of November

"What about Power Pages?"

PUBLIC PREVIEW

FOUR SITE-LEVEL METRICS

USAGE

Session count

ENGAGEMENT

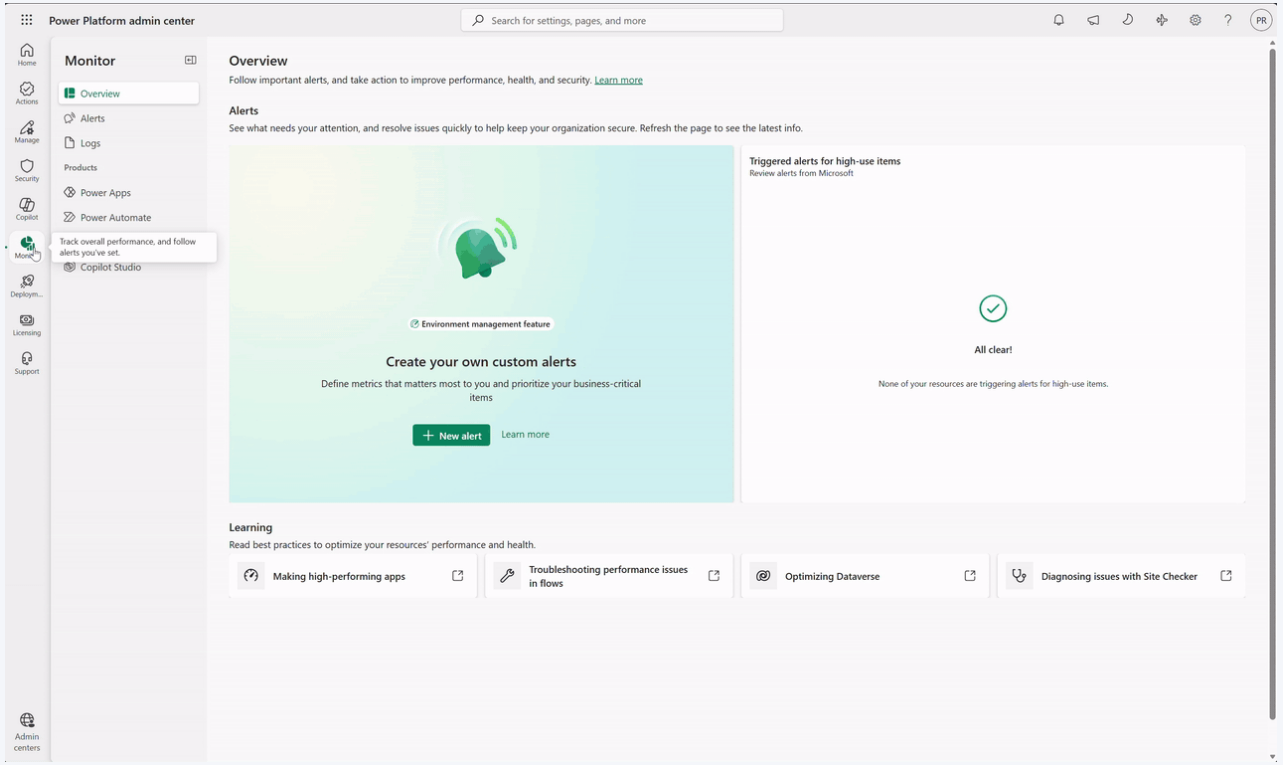
Average session duration

PERFORMANCE

Page load time (P75)

RELIABILITY

Error rate



Next: more site/page metrics + alerts

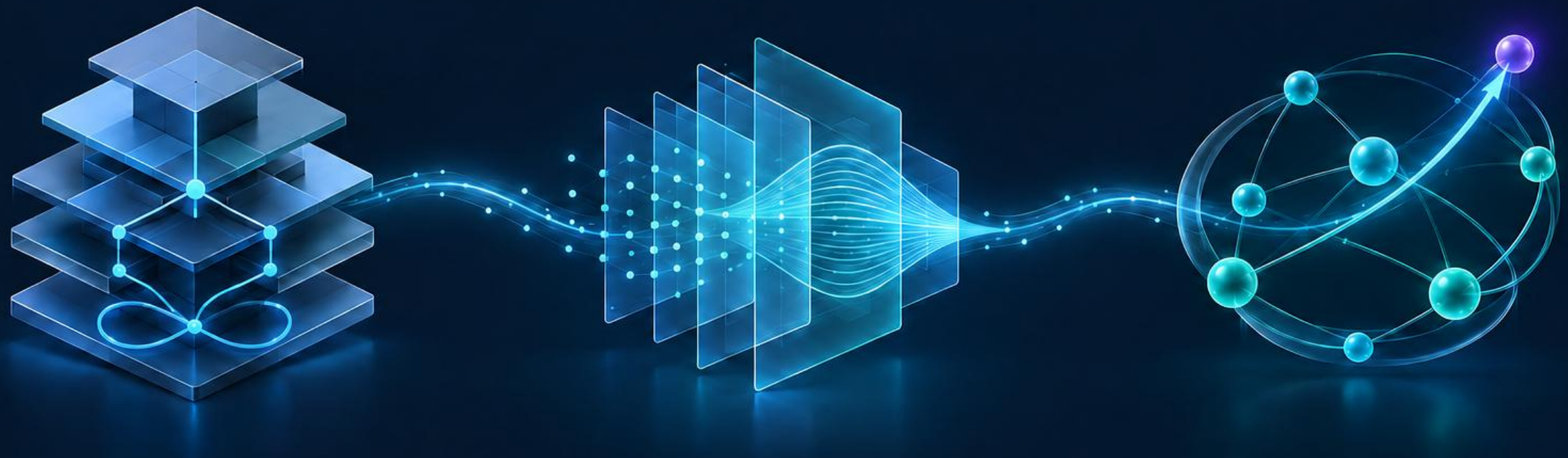
Monitor support today

✓ General support — Not currently available | Hourly eligibility varies

Product	Resource types	Metrics	Recommendations	Alerts	Where to see metrics/recs/alerts
Copilot Managed Runtime	Apps	✓ Public preview	—	✓ Public preview	**Microsoft admin center (all)**
Power Apps	Code apps Model-driven apps Canvas apps	✓	✓	✓	**PPAC (all) & make.powerapps.com/monitor (metrics/recs)**
Power Automate	Cloud flows Desktop flows	✓	✓	✓	**PPAC (all) & Automation Center (metrics)**
Copilot Studio	Conversational agents Autonomous agents	✓	✓	✓	**PPAC (all) & Monitor tab in Copilot Studio (metrics)**
Power Pages	Sites	✓ Public preview	—	—	**PPAC (metrics)**

Coming soon: Copilot Managed Runtime Agents and Power Platform Agent Flows

The future of DevOps



ESTABLISHED

Foundation

Git · Deploy · Extend · Recover · Monitor

Intelligence

Signals into insight

Quality · Reliability · Efficiency

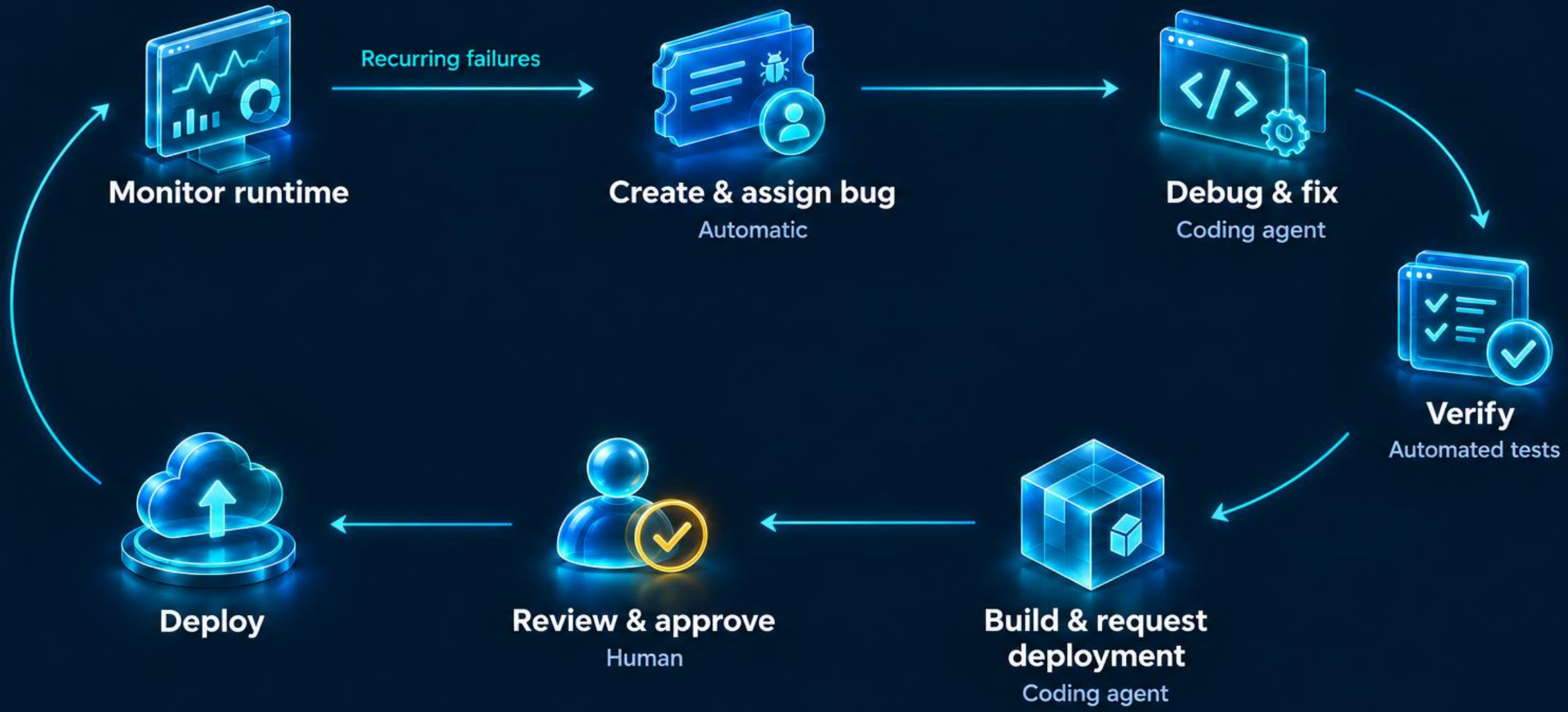
NEXT FRONTIER

Autonomy

Specialized AI agents

Autonomous. Self-improving.

Intelligent, autonomous operations



Demo:
Intelligent, Autonomous Ops

Early look

Thank You!

Related upcoming sessions

TUE · OCT 27
2:00 PM – 3:00 PM
Boulevard Ballroom 162

What's New and What's Next in the Managed Platform
Austin Laugesen · David Blyth · Mihaela Blendea · Zohar Raz

WED · OCT 28
8:00 AM – 9:00 AM
Premier BR 319

Managed Security: Securing and Governing Apps, Automations, and Agents at Scale
Gregg Barker · Lane Swenka

WED · OCT 28
10:00 AM – 11:00 AM
Boulevard Ballroom 162

Governing Without Stifling Innovation
Ken Auguillard · Mik Ferland · Sid Gundavarapu

THU · OCT 29
10:45 AM – 11:45 AM
Grand Ballroom 119

Take Control of Your Power Platform Spend
Amiya Patra · Graham Mosley

THU · OCT 29
12:00 PM – 1:00 PM
Chairman's BR 370

From Laptop to Governed Production: Creating and Running a Hosted Managed Agent
Matthew Barbour

THU · OCT 29
12:00 PM – 1:00 PM
Boulevard Ballroom 163

Build Fast, Govern Better: The Latest in Power Apps code apps and Copilot Managed Runtime for devs
Alan Castellanos Moreno · Austin Laugesen · Jordan Chodak · Ryan Jansen

POWER PLATFORM COMMUNITY CONFERENCE

Session Feedback Surveys

We really want to hear from YOU!

In the pursuit of making next year's Power Platform Community Conference even better, we want to hear your feedback about this session.

Here's How -

- *Simply go to the Whova App on your smartphone*
- *Scroll down on the Power Platform Community Conference Homepage to 'Additional Resources' to click "Surveys'.*
- *Click Session Feedback.*
- *Scroll down to find this session title.*
- *Complete the session feedback survey.*
- *Finally, click 'Submit'*

It's just that easy!

